

Q2)

Apply all CPU scheduling algorithms for insem exam of OS.

<u>Process</u>	<u>AT</u>	<u>BT</u>	<u>Priority</u>
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

FCFS (First come first serve)

Ready Queue (RQ)

0 :-

P ₁	P ₂	P ₃	P ₄	P ₅	P ₆
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1 :-

P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈
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2 :-

P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
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4 :-

P₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
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13 :-

P₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
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19 :-

P₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
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23 :-

P₄	P ₅	P ₆	P ₇	P ₈	P ₉
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32 :-

P₅	P ₆	P ₇	P ₈	P ₉
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38 :-

P₆	P ₇	P ₈	P ₉
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42 :-

P₇	P ₈	P ₉
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51 :-

P₈	P ₉
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57 :-

P₉

Gantt chart

P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	
0	4	13	19	23	32	38	42	51	57

<u>Process</u>	<u>AT</u>	<u>BT</u>	<u>CT</u>	<u>TAT</u>	<u>WT</u>	<u>RT</u>
P ₁	0	4	4	4	0	0
P ₂	0	9	13	13	4	4
P ₃	0	6	19	19	13	13
P ₄	0	4	23	23	19	19
P ₅	0	9	32	32	23	23
P ₆	0	6	38	38	32	32
P ₇	1	4	42	41	37	37
P ₈	1	9	51	50	41	41
P ₉	2	6	57	55	49	49

Avg waiting time

$$\frac{0 + 4 + 13 + 19 + 23 + 32 + 37 + 41 + 49}{9} = \underline{\underline{24.22}}$$

Avg TurnAround time

$$\frac{4 + 13 + 19 + 23 + 32 + 38 + 41 + 50 + 55}{9} = \underline{\underline{30.56}}$$

$$\underline{\underline{\text{Avg Remaining time} \Rightarrow 24.22}}$$

SJF (shortest - job - first)

0:-

P_1

 $\leftarrow P_1$ is selected (BT=4)

4:-

P_1	P_4
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 $\leftarrow P_1$ completed, P_4 selected (BT=4)

8:-

P_4	P_7
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 $\leftarrow P_4$ completed, P_7 selected (BT=4)

12:-

P_7	P_3
-----------------------------	-------

 $\leftarrow P_7$ completed, P_3 selected (BT=6)

18:-

P_3	P_6
-----------------------------	-------

 $\leftarrow P_3$ completed, P_6 selected (BT=6)

24:-

P_6	P_9
-----------------------------	-------

 $\leftarrow P_6$ completed, P_9 selected (BT=6)

30:-

P_9	P_2
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 $\leftarrow P_9$ completed, P_2 selected (BT=9)

39:-

P_2	P_5
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 $\leftarrow P_2$ completed, P_5 selected (BT=9)

48:-

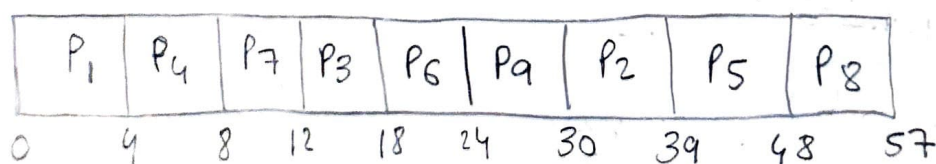
P_5	P_8
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 $\leftarrow P_5$ completed, P_8 selected (BT=9)

57:-

P_8

Gantt chart



<u>Process</u>	<u>AT</u>	<u>BT</u>	<u>CT</u>	<u>TAT</u>	<u>WT</u>	<u>RT</u>
P ₁	0	4	4	4	0	0
P ₂	0	9	39	39	30	30
P ₃	0	6	18	18	12	12
P ₄	0	4	8	8	4	4
P ₅	0	9	48	48	39	39
P ₆	0	6	24	24	18	18
P ₇	1	4	12	11	7	7
P ₈	1	9	57	56	47	47
P ₉	2	6	36	28	22	22

$$WT = RT$$

$$\begin{aligned} \text{CPU length} &= 57 - 0 \\ &= 57 \text{ units} \\ &= \end{aligned}$$

$$\text{Throughput} = \frac{\text{No. of Processes completed}}{\text{Total CPU time}} = \frac{9}{57}$$

$$= 0.158$$

$$\text{Avg WT} = 19.89$$

$$\text{Avg TAT} = 26.22$$

$$\text{Avg RT} = 19.89$$

SRTF (Shortest Remaining time First)

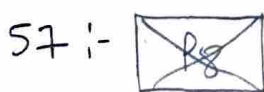
RQ

0:- $\boxed{P_1}$ $\in P_1$ has shortest remaining time

$$P_1 = 4 - 1 = 3$$



$$P_4 = 4, P_7 = 4, P_3 = 6, P_9 = 6, P_2 = 9, P_5 = 9, P_8 = 9, P_6 = 6$$



GT

P_1	P_4	P_7	P_3	P_6	P_9	P_2	P_5	P_8	
0	4	8	12	18	24	30	39	48	57

<u>Process</u>	<u>AT</u>	<u>BT</u>	<u>CT</u>	<u>TAT</u>	<u>WT</u>	<u>RT</u>
P ₁	0	4	4	4	0	0
P ₂	0	9	39	39	30	30
P ₃	0	6	18	18	12	12
P ₄	0	4	8	8	4	4
P ₅	0	9	48	48	39	39
P ₆	0	6	24	24	18	18
P ₇	1	4	12	11	7	7
P ₈	1	9	57	56	47	47
P ₉	2	6	30	28	22	22

$$\underline{\text{Avg WT}} := \frac{179}{9} = 19.89$$

$$\underline{\text{Avg TAT}} := \frac{236}{9} = 26.22$$

$$\underline{\text{Avg RT}} := 19.89$$

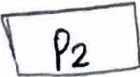
$$\underline{\text{CPU Length}} := 4 + 9 + 6 + 4 + 9 + 6 + 4 + 9 + 6 = \underline{\underline{57 \text{ units}}}$$

$$\underline{\text{Throughput}} := \frac{\text{No. of Processes}}{\text{Total time}} = \frac{9}{57} = 0.158$$

Priority Scheduling - Non Preemptive

<u>Process</u>	<u>AT</u>	<u>BT</u>	<u>Priority</u>
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

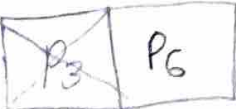
RQ

0 :-  P₂ = P₅ = Priority 1 → P₂ selected

9 :-  P₅ = P₈ = Priority 1 → P₅ selected

18 :-  Priority 1 → P₈ selected

27 :-  P₃ = P₆ = P₉ = Priority 2 → P₃ selected

33 :-  P₆ = P₉ = Priority 2 → P₆ selected

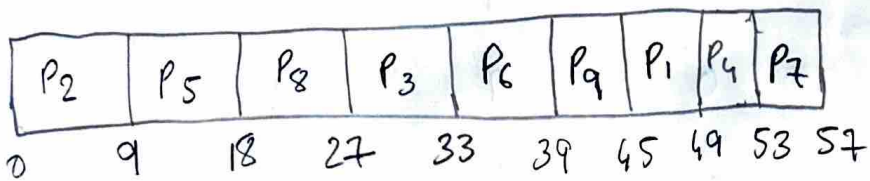
39 :- $\boxed{\cancel{P_6} \quad P_9}$ $\rightarrow P_9$ selected

45 :- $\boxed{\cancel{P_9} \quad P_4}$ $\rightarrow P_4 = P_7 = \text{Priority 3} \rightarrow P_4$ selected

49 :- $\boxed{\cancel{P_4} \quad P_7}$

53 :- $\boxed{\cancel{P_7}}$ $\rightarrow$ Done

Gantt chart



<u>Process</u>	<u>AT</u>	<u>BT</u>	<u>CT</u>	<u>TAT</u>	<u>WT</u>	<u>RT</u>
P_1	0	4	49	49	45	45
P_2	0	9	9	9	0	0
P_3	0	6	33	33	27	27
P_4	0	4	53	53	49	49
P_5	0	9	18	18	9	9
P_6	0	6	39	39	33	33
P_7	1	4	57	56	52	52
P_8	1	9	27	26	17	17
P_9	2	6	45	43	37	37

Avg WT :- $\frac{29.89}{2}$

Avg RT :- $\frac{29.89}{2}$

Throughput :- $\frac{9}{57}$

Avg TAT :- $\frac{36.22}{2}$

CPU length :- $\frac{57 \text{ units}}{2}$

$\frac{= 0.158}{2}$

Priority scheduling $\rightarrow$ Preemptive

<u>process</u>	<u>AT</u>	<u>BT</u>	<u>Priority</u>
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

RQ

0 :- $\boxed{P_2} \rightarrow P_2 = P_5 = 1 \rightarrow P_2$ selected

1 :- $\boxed{P_2} \rightarrow P_2 = 1, P_8 = 1 \rightarrow$ equal Priority $\rightarrow P_2$ continues

2 :- $\boxed{P_2} \rightarrow P_2 = 1 < P_9 = 2 \rightarrow P_2$ continues

9 :- $\boxed{\cancel{P_2}} \boxed{P_5} \rightarrow P_5 = P_8 = 1 \rightarrow P_5$ selected

18 :- $\boxed{\cancel{P_5}} \boxed{P_8} \rightarrow P_8 = 1 \rightarrow$ highest

27 :- $\boxed{\cancel{P_8}} \boxed{P_3} \rightarrow P_3 = P_6 = P_9 = 2 \rightarrow P_3$ selected

33 :-

P₃	P ₆
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 $\rightarrow P_6 = P_9 = 2 \rightarrow P_6$ selected.

39 :-

P₆	P ₉
--------------------------	----------------

 $\rightarrow P_9$ selected

45 :-

P₉	P ₁
--------------------------	----------------

 $\rightarrow P_1 = P_4 = P_7 = 3 \rightarrow P_1$ selected

49 :-

P₁	P ₄
--------------------------	----------------

 $\rightarrow P_4 = P_7 = 3 \rightarrow P_4$ selected

53 :-

P₄	P ₇
--------------------------	----------------

57 :-

P₇

GT

P_2	P_5	P_8	P_3	P_6	P_9	P_1	P_4	P_7	
0	9	18	27	33	39	45	49	53	57

CT TAT WT RT

49 49 45 45

9 9 0 0

33 33 27 27

53 53 49 49

18 18 9 9

39 39 33 33

57 56 52 52

27 26 17 17

45 43 37 37

$$\text{Avg WT} := \frac{269}{9} = 29.89$$

$$\text{Avg TAT} := \frac{326}{9} = 36.22$$

$$\text{Avg RT} := 29.89$$

CPU length :- 57 units

ThroughPut :- 0.158